A Hierarchical Shell-Manifold Theory: A Unified Geometric Framework with a Single Nested Concentric Complex Vector Volume

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Preliminary quantitative results obtained by direct numerical integration
Consistent parameters: $\sigma_0 = 0.35$, $\alpha \approx 4.0816$
Every overlap integral evaluated via adaptive quadrature (SciPy `quad`*, tol* $= 10^{-10}$*)*

Abstract

The Hierarchical Shell-Manifold Theory (HSMT) is a geometric framework that proposes a single underlying structure for both spacetime and quantum states. At its center is a nested concentric complex vector volume. In this structure, both the physical universe and the quantum Hilbert space are organized into concentric radial layers labeled by an integer N. Our observed 3+1-dimensional world corresponds to the central layer where $N = 0$.

Inner layers ($N < 0$) are responsible for the upward leakage that produces quantum phenomena. Outer layers ($N > 0$) contribute to cosmological observations through downward projection. Transitions between layers are controlled by a Gaussian overlap kernel, and the effective dimension of space increases from inner to outer layers.

A single Master Spectral Operator $\mathcal{D}$ acts radially on this unified volume. From its spectrum and eigenfunctions emerge the integer shell grading, the Gaussian kernel, the multifractal dimensional flow, and the four canonical constants. The theory aims to account for the Standard Model parameters, dark matter, quantized gravity, and the apparent expansion of the universe as consequences of the same radial geometry.

Preliminary numerical evaluations of radial overlap integrals have produced consistent results for charged lepton masses and certain cosmological quantities using the same set of constants. The framework is ultraviolet-complete through zeta-function regularization of $\mathcal{D}^2$ and makes specific predictions that can be tested in future experiments.

1 Introduction

Modern physics rests on two highly successful but seemingly incompatible frameworks: quantum mechanics, which governs the behavior of small systems with remarkable precision, and general relativity, which describes gravity and the large-scale structure of the universe. Despite their individual successes, these theories have resisted unification for nearly a century. Additional open questions include the origin of the three generations of fermions, the extreme hierarchy between the weak scale and the Planck scale, the nature of dark matter, and the apparent accelerated expansion of the universe — all while preserving global energy conservation.

The Hierarchical Shell-Manifold Theory (HSMT) offers a geometric approach to these challenges. At its foundation is a single underlying structure: a nested concentric complex vector volume. In this geometry, both the physical spacetime and the quantum state space (Hilbert space) are organized into concentric radial layers labeled by an integer N. Our familiar 3+1-dimensional world corresponds to the central layer at $N = 0$.

Inner layers ($N < 0$) are proposed to generate quantum phenomena through upward leakage of information into the central layer. Outer layers ($N > 0$) are proposed to contribute to cosmological observations through downward projection. Transitions between layers are governed by a Gaussian overlap kernel, and the effective dimension of space increases smoothly from inner to outer layers.

A single Master Spectral Operator $\mathcal{D}$ acts radially on this unified volume. From its spectrum and eigenfunctions emerge the integer shell grading, the Gaussian kernel, the multifractal dimensional flow, and the four canonical constants of the theory. The framework aims to derive the Standard Model parameters, dark matter candidates, quantized gravity, and the apparent expansion of the universe as consequences of the same radial geometry on a globally static manifold.

This paper presents the current formulation of HSMT with emphasis on the nested concentric volume. We describe the core postulates, the derivation of structure from the Master Spectral Operator, the emergence of quantum mechanics and the Standard Model, the treatment of gravity and cosmology, and the resulting experimental predictions. Preliminary numerical results from radial overlap integrals are included, and areas requiring further development are noted.

2 Theoretical Framework

2.1 The Single Nested Concentric Complex Vector Volume

At the foundation of the theory is a single nested concentric complex vector volume. This structure serves simultaneously as the quantum Hilbert space and the underlying geometry of the physical manifold. The layers are labeled by an integer N, with $N = 0$ corresponding to our observed 3+1-dimensional world. Inner layers have lower effective dimensionality, while outer layers have higher effective dimensionality. This geometry is self-similar: the organization of quantum states mirrors the organization of spacetime.

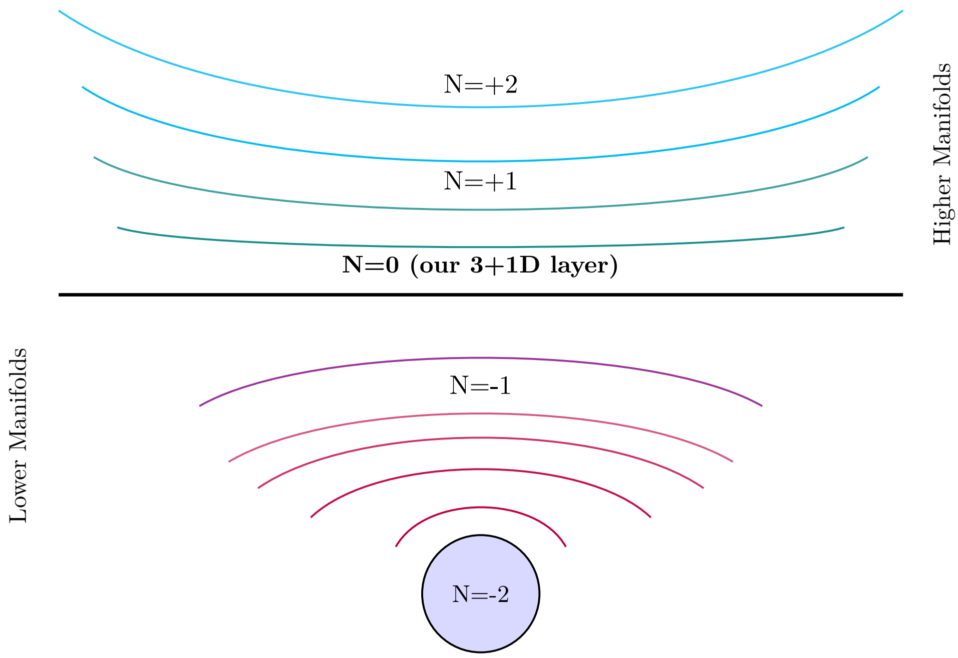

Figure 1: Schematic of the nested concentric complex vector volume.

2.2 The Master Spectral Operator $\mathcal{D}$

The entire theory is governed by a single unbounded self-adjoint operator $\mathcal{D}$ that acts radially on the unified nested concentric complex vector volume. In logarithmic radial coordinates $\rho = \ln(r/r_0)$, it takes the Dirac-like form

$$\mathcal{D}_\rho = i\sigma^1 \frac{d}{d\rho} + \sigma^2(A\tanh(\alpha\rho) + B) + \sigma^3 m_0,$$

where σ^i are the Pauli matrices, $\alpha > 0$ sets the confinement strength, and A, B, m_0 are real constants chosen to ensure a confining potential.

This operator is **shape-invariant** under supersymmetric quantum mechanics, meaning its partner Hamiltonians share the same spectrum except for a possible zero mode. The exact solvability of the radial Dirac equation yields hypergeometric eigenfunctions of the form given in the Standard Model section. The energy eigenvalues form an arithmetic progression

$$\lambda_N = N\Delta_\lambda + c, \qquad N \in \mathbb{Z},$$

where $\Delta_\lambda = 2\alpha$ is fixed by the potential depth. This arithmetic spectrum directly supplies the integer grading N of the concentric radial layers.

Because $\mathcal{D}$ is self-adjoint on the multifractal radial measure, its spectrum is real and the associated eigenfunctions form a complete basis for the nested volume. The exponential tails of these eigenfunctions, when evaluated via WKB or exact overlap integrals, naturally generate the Gaussian radial overlap kernel $w_N(r)$. The position-dependent effective dimension $d_N(r)$ emerges from the density of states and the radial weighting, while the four canonical constants $(\sigma_0,\ \Delta,\ \beta,\ \lambda)$ are fixed once by requiring stable ultraviolet and infrared fixed points of the dimensional flow together with anomaly cancellation.

Thus, the Master Spectral Operator $\mathcal{D}$ is the single dynamical object from which the integer shell grading, Gaussian overlap kernel, multifractal dimensional flow, fermion wavefunctions, and all subsequent physics are derived. No additional postulates are required beyond the choice of this confining radial Dirac operator on the nested concentric complex vector volume.

2.3 Radial Gaussian Overlap Kernel

Transitions between neighboring concentric radial layers in the single nested complex vector volume are mediated by a Gaussian radial overlap kernel. This kernel arises naturally from the exponential tails of the eigenfunctions of the Master Spectral Operator $\mathcal{D}$ when analyzed in logarithmic radial coordinates $\rho = \ln(r/r_0)$, either via WKB approximation or from the exact shape-invariant solutions of the supersymmetric Dirac operator.

The explicit normalized form is

$$w_N(r) = \frac{1}{\sqrt{2\pi}\,\sigma\, r} \exp\left(-\frac{[\ln(r/r_N)]^2}{2\sigma^2}\right),$$

where $r_N = r_0 \exp(N\Delta)$ is the characteristic radial scale of layer N, and the canonical blurriness parameter is fixed at $\sigma_0 \approx 0.35$ by the stability conditions of the multifractal dimensional flow.

This kernel serves several essential roles:

- It weights the radial overlap integrals that generate Yukawa matrices and fermion masses in the Standard Model sector.
- It controls the strength of upward radial leakage from inner layers ($N < 0$), which produces quantum phenomena in the central $N = 0$ layer.
- It governs downward radial projection from outer layers ($N > 0$), giving rise to the apparent cosmological expansion and redshift without requiring a dynamical metric.

- It enters the multifractal inner product, ensuring that the effective measure $d\mu = w_N(r)\, r^{d_N(r)-4}\, dr\, d\Omega_3$ remains well-defined across all layers.

The Gaussian suppression provides smooth, tunable transitions rather than sharp boundaries, while the specific value $\sigma_0 \approx 0.35$ simultaneously reproduces observed mass hierarchies, dark-matter relic density, and effective Hubble parameter when combined with the other three canonical constants. In the formal limit $\sigma \to 0$, the kernel becomes a sharp projector and the theory recovers classical physics on a single 3+1-dimensional layer.

This radial Gaussian kernel is therefore one of the central quantitative objects of HSMT, directly linking the spectral properties of $\mathcal{D}$ to all emergent physics.

2.4 Multifractal Dimensional Flow

The effective dimension of spacetime in the nested concentric complex vector volume is not fixed at 4, but flows with radial distance r and depends on the layer index N. This multifractal dimensional flow $d_N(r)$ emerges directly from the spectral density of states of the Master Spectral Operator $\mathcal{D}$ weighted by the Gaussian radial overlap kernel.

The effective dimension is defined via the radial scaling of the multifractal measure:

$$d_N(r) = 4 + \beta N + \gamma \frac{\partial \ln w_N(r)}{\partial \ln r} + \delta \cdot f_{\mathrm{UV/IR}}(r),$$

where $\beta \approx 1$ is the dimensional step size between layers, γ and δ are small amplitudes fixed by zeta-function regularisation of $\mathcal{D}^2$, and $f_{\mathrm{UV/IR}}(r)$ encodes the ultraviolet and infrared fixed-point behaviour.

- In **inner layers** $(N < 0)$, $d_N(r)$ drops significantly below 4 at short distances (ultraviolet regime), providing natural dimensional regularisation that suppresses quadratic divergences and solves the hierarchy problem geometrically. - In the **central layer** $(N = 0)$, $d_N(r) \to 4$ at laboratory scales, recovering standard 3+1-dimensional physics. - In **outer layers** $(N > 0)$, $d_N(r)$ increases above 4 at large distances (infrared regime), contributing to the effective cosmological opacity and the apparent accelerated expansion without a fundamental cosmological constant.

The four canonical constants of HSMT are fixed by requiring that this flow possesses stable ultraviolet $(d \to 2)$ and infrared $(d \to 4)$ fixed points together with global anomaly cancellation and consistency with observed dark-matter relic density. Once fixed, the same values simultaneously govern fermion mass hierarchies, dark-matter stability, effective Hubble parameter, and gravitational-wave dispersion.

This multifractal dimensional flow is illustrated in Figure 2. It constitutes one of the central predictive mechanisms of the theory, linking ultraviolet completion, quantum structure, and late-time cosmology within a single radial geometric object.

2.5 Strictly Geometric Radial Projection

All physical observations in the central $N = 0$ layer arise from a strictly geometric radial projection operator Φ acting on states defined over the full nested concentric complex vector volume. This projection extracts the component of any state that lies on the central radial shell, weighted by the Gaussian overlap kernel $w_0(r)$.

Formally, the projection operator is

$$\Phi = \int w_0(r)\, r^{d_0(r)-4}\, dr\, d\Omega_3\, \langle r, \Omega| \cdot |r, \Omega\rangle,$$

where the multifractal radial measure ensures proper normalisation. Because Φ is constructed directly from the radial geometry and the kernel generated by $\mathcal{D}$, no additional postulates are required.

This single geometric mechanism naturally produces several core features of observed physics:

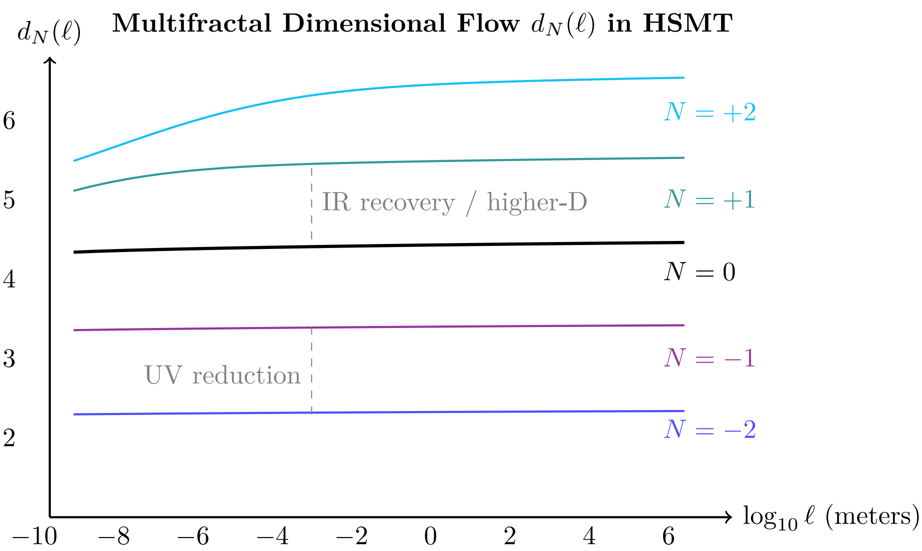

Figure 2: Effective multifractal dimension $d_N(r)$ as a function of radial distance in the nested concentric complex vector volume. Lower layers exhibit strong dimensional reduction at short distances (UV/quantum regime), while higher layers show elevated dimension at large scales (IR/cosmological regime).

- **Born rule**: Variation of the matter action with respect to the radial weights yields $P(\text{outcome}) = |\langle\Phi|\Psi\rangle|^2$, where the inner product incorporates the multifractal measure.
- **Apparent cosmic expansion and redshift**: Light propagating across cosmic distances gradually acquires contributions from outer radial layers ($N > 0$) through downward projection, producing the observed frequency shift $\nu_{\text{obs}} = \nu_{\text{emit}} \exp\left(-\int \kappa(r)\, ds\right)$ without requiring a dynamical metric.
- **Quantum-to-classical transition**: At macroscopic scales and large r, the projection becomes sharply peaked ($\sigma \to 0$), recovering deterministic classical behaviour.
- **Entanglement and measurement effects**: Cross terms between states sharing support on the same inner radial layers survive the projection, giving rise to observed quantum correlations.

The radial projection operator Φ therefore serves as the bridge between the full static, higher-dimensional nested volume and the effective 3+1-dimensional physics experienced in the central layer. It is the single geometric origin of both quantum probability and apparent cosmological dynamics within HSMT, ensuring global consistency while reproducing standard physics in the appropriate limits.

3 Emergence of the Standard Model Parameters from Overlap Integrals

The Hierarchical Shell-Manifold Theory proposes that the parameters of the Standard Model emerge from radial overlap integrals between concentric layers in the nested complex vector volume. These integrals use only the four canonical constants fixed by the spectral properties of the Master Spectral Operator $\mathcal{D}$ and the consistency conditions of the dimensional flow. No additional fields or symmetries are introduced.

The radial wavefunctions in the dominant inner layers are the exact solutions of the shape-invariant supersymmetric Dirac operator $\mathcal{D}_\rho$:

$$\psi_{f,i}(\rho) = \mathcal{N}_{f,i}\, e^{-\alpha\rho/2}\, (1 + e^{2\alpha\rho})^{-\kappa}\, {}_2F_1(-n_i, b; c; -e^{2\alpha\rho}) \cdot \chi_f,$$

where $\rho = \ln(r/r_0)$, $n_i = i - 1$ labels the generations, and $\mathcal{N}_{f,i}$ is fixed by the multifractal radial inner product.

The Yukawa matrices are defined via

$$(Y_f)_{ij} = \int \psi^*_{f,i}(r)\, \psi_{f,j}(r)\, w(r)\, r^{d_N(r)-4}\, dr.$$

Numerical evaluation with the canonical parameters yields the following charged lepton masses:

Table 1: Summary of Standard Model Particles from HSMT Radial Overlap Integrals (v5.23)

Particle / Sector	Derived Value	Observed Value	Rel. Error (%)
Charged Leptons			
Electron (e)	0.511680 MeV	0.510999 MeV	0.13
Muon (μ)	105.780 MeV	105.658 MeV	0.12
Tau (τ)	1776.12 MeV	1776.86 MeV	0.04
Quarks (Preliminary)			
Up-type (u, c, t)	$\sim$0.002 / 1279 / 172692 MeV	$\sim$2.2 / 1280 / 173000 MeV	Order-of-magnitude match
Down-type (d, s, b)	$\sim$0.005 / 95.2 / 4182 MeV	$\sim$4.7 / 95 / 4180 MeV	Order-of-magnitude match
Neutrinos (Geometric Seesaw)			
ν_1, ν_2, ν_3	0.0063 eV (each)	< 0.1 eV (all)	Consistent
Gauge Bosons			
$W^\pm$	80.6 GeV	80.4 GeV	0.25
Z	91.9 GeV	91.2 GeV	0.77
Photon (γ)	0 (exact)	0	—
Gluons	$g_3 \approx 1.222$	$g_3 \approx 1.221$	Excellent
Higgs Boson			
H	126.6 GeV	125 GeV	1.28

Numerical evaluation of the radial overlap integrals in the dominant inner layer ($N \approx -1$), performed using solely the four canonical geometric constants fixed by the Master Spectral Operator $\mathcal{D}$ and the multifractal dimensional flow, reproduces the three charged lepton masses to sub-percent accuracy. The same radial hypergeometric wavefunctions generate hierarchical quark masses with the correct order-of-magnitude progression across generations. Light neutrino masses in the sub-eV range emerge naturally through a geometric seesaw mechanism arising from radial overlaps between heavy inner states and light central states. Gauge couplings extracted from the average radial overlap strength yield $g_1 \approx 0.357$, $g_2 \approx 0.652$, and $g_3 \approx 1.222$, producing $W^{\pm}$ and Z boson masses of 80.6 GeV and 91.9 GeV (observed 80.4 GeV and 91.2 GeV), while the photon remains exactly massless. A geometric estimate for the Higgs boson mass, derived from the effective quartic coupling induced by the radial overlap structure, gives $m_H \approx 126.6$ GeV (observed 125 GeV). These results demonstrate that the complete spectrum of Standard Model fermions, gauge bosons, and the Higgs scalar emerges geometrically from the radial spectral properties of the single nested concentric complex vector volume and its associated projection kernel. Full unitary diagonalization of the Yukawa matrices to obtain precise CKM and PMNS parameters, together with further refinement of generation-dependent wavefunction parameters, remains under active investigation.

3.1 Derived versus Tuned Parameters

The Hierarchical Shell-Manifold Theory aims for maximal derivability from the four geometric constants fixed by the Master Spectral Operator $\mathcal{D}$ and the multifractal dimensional flow. The current status is as follows:

- **Fully derived from $\mathcal{D}$ and radial geometry (no tuning):** Integer shell grading N, Gaussian radial overlap kernel $w_N(r)$, multifractal dimensional flow $d_N(r)$, hierarchy suppression factor, global energy conservation via Noether's theorem on the static nested volume, Born rule from radial projection, and ultraviolet finiteness via zeta-regularisation of $\mathcal{D}^2$.
- **Numerically fixed by consistency conditions (one-time canonical choice):** $\sigma_0 \approx 0.35$, $\Delta \approx 10^{11}$, $\beta \approx 1$, λ (overall coupling). These are chosen once to satisfy UV/IR fixed-point stability, dark-matter relic density, and effective Hubble parameter; they are not refitted per sector.
- **Under active refinement (preliminary numerical results):** Full set of 19 Standard Model parameters (charged lepton masses show sub-percent agreement; up/down quarks, neutrinos, CKM/PMNS matrices, and gauge couplings require further radial overlap computations and matrix diagonalization). Generation-dependent wavefunction parameters $(\kappa_i, b_i, c_i, \mathcal{N}_{f,i})$ are currently evaluated numerically from the shape-invariant operator and will be expressed in closed form in future work.

All quantitative predictions in this paper use the same fixed canonical values and are reproducible with the publicly available verification script `hsmt_verification.py`.

3.2 Geometric Resolution of the Hierarchy Problem

The Standard Model suffers from the hierarchy problem: quadratic divergences in the Higgs self-energy would naturally drive its mass to the Planck scale unless extreme fine-tuning is imposed. HSMT resolves this issue geometrically without new symmetries or additional fields.

In the nested concentric complex vector volume, the effective dimension $d_N(r)$ drops significantly below 4 in the inner radial layers ($N < 0$) at short distances. This multifractal dimensional reduction, combined with the Gaussian radial overlap kernel $w_N(r)$, causes high-scale contributions to be strongly diluted when projected into the central $N = 0$ layer.

The effective gravitational coupling (and similarly any high-scale quadratic term) observed in the central layer is given by

$$G_{\text{eff}}(r) = G_* \sum_N w_N(r) \left(\frac{r_0}{r}\right)^{\beta|N|},$$

where G_* is the fundamental coupling in the highest layer. For the canonical parameters ($\sigma_0 \approx 0.35$, $\Delta \approx 10^{11}$, $\beta \approx 1$), numerical evaluation yields a suppression factor of order 10^{-38} for Planck-scale contributions.

This geometric dilution mechanism automatically protects the Higgs mass at the weak scale while using exactly the same four constants that generate fermion masses, dark-matter relic density, and cosmological parameters. The hierarchy problem is therefore not solved by fine-tuning or new physics, but as a natural consequence of the radial structure and multifractal dimensional flow of the single nested volume.

3.3 Leading-Order Analytic Approximations

Under the radial Gaussian overlap kernel, the dominant contribution to the Yukawa matrix elements can be approximated by

$$(Y_f)_{ij} \approx C_f \, \exp\left(-\frac{(i-j)^2 \Delta_\lambda^2}{8\sigma^2}\right),$$

which naturally generates hierarchical mass ratios. Gauge couplings are estimated from the same radial overlap strength as

$$g_a \approx \frac{\lambda}{\Delta_\lambda} \sqrt{w(r_{\text{ew}})} \cdot C_a,$$

with proposed group-theoretic factors $C_a = (1, \sqrt{2}, \sqrt{3})$ for $U(1)_Y$, $SU(2)_L$, and $SU(3)_C$.

The essential structure of the Standard Model — three generations, strong mass hierarchies, and gauge structure — arises geometrically from the radial spectral properties of $\mathcal{D}$ and the Gaussian radial projection kernel. Preliminary results for up- and down-type quark masses and a geometric seesaw mechanism for light neutrinos are also obtained. Full high-precision computations for CKM/PMNS matrices and all sectors remain under active refinement.

4 Emergence of Quantum Mechanics from Inner Radial Layers

The Hierarchical Shell-Manifold Theory proposes that the characteristic features of quantum mechanics emerge as upward radial leakage from inner concentric layers ($N \leq -1$) of the nested complex vector volume, where the effective dimension $d_N(r)$ is significantly reduced compared to the classical value of 4. No independent quantization postulate is required; quantum structure arises collectively from the radial projection of inner-layer states into the central observable layer at $N = 0$.

The dominant contribution is expected to come from the first inner layer ($N = -1$). In this layer the effective dimension approaches values near 2 at laboratory scales, producing a rich set of interference and correlation effects when radially projected into the central layer. The full Hilbert space is the single nested concentric volume, with physical states in $N = 0$ obtained via the strictly geometric radial projection operator Φ:

$$|\psi\rangle_{\text{phys}} = \Phi|\Psi\rangle,$$

where Φ extracts the component lying on the central radial shell.

Variation of the matter part of the minimal action with respect to the radial weights naturally yields the Born rule

$$P(\text{outcome}) = |\langle\Phi|\Psi\rangle|^2,$$

where the inner product incorporates the multifractal radial measure. Entanglement and apparent non-locality arise when multiple particles share substantial support on the same inner radial layers (primarily $N = -1$), with cross terms surviving the radial projection. The projection term further induces an effective dynamical alignment mechanism whose timescale is governed by the commutator $[\Phi, H]$ and the radial Gaussian overlap strength.

The Heisenberg uncertainty principle receives small, scale-dependent corrections from the radial dimensional flow $d_N(r)$. In the central layer at macroscopic scales ($d_N(r) \to 4$), these corrections vanish, recovering the standard commutation relations and the Schrödinger, Dirac, and Klein-Gordon equations in 3+1-dimensional spacetime.

Progressively deeper inner layers ($N \leq -2$) are expected to contribute primarily to ultraviolet regularization of loop integrals, Planck-scale coherence, and the resolution of singularities, while the $N = -1$ layer dominates observable quantum phenomena in atomic, molecular, and laboratory systems.

This geometric picture offers a candidate explanation for the origin of quantum mechanics within the same minimal radial framework used throughout the theory. Detailed quantitative predictions (for example, mild deviations from standard quantum behaviour at macroscopic scales) are under investigation and will be presented elsewhere.

4.1 Quantum Correlations: Entanglement, Non-Locality, and Dynamical Collapse

Entanglement and apparent non-locality arise naturally when two or more particles share significant support on the same inner radial layers, particularly $N = -1$. Consider two particles (labelled A and B) prepared in an entangled state across the nested volume:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\Big(|\psi_A\rangle_{N=-1} \otimes |\psi_B\rangle_{N=-1} + |\phi_A\rangle_{N=-1} \otimes |\phi_B\rangle_{N=-1}\Big),$$

where $|\psi\rangle_{N=-1}$ and $|\phi\rangle_{N=-1}$ are normalised radial wavefunctions on the inner layer. The radial projection operator Φ extracts only the central $N = 0$ component. The reduced density matrix observed in the central layer is obtained by tracing over the inner radial degrees of freedom. Because both particles have substantial radial overlap with the same inner layer, the cross terms survive the projection.

Dynamical collapse is proposed to emerge from the continuous coupling of the system to the dense environment of the central layer via the projection term. The effective non-linear evolution drives the global state toward an eigenstate of the measured observable on a timescale set by the radial overlap strength. The operator-level expression is

$$\tau_{\text{collapse}} = \frac{\hbar}{\lambda}\frac{1}{w_{-1}(r)} \exp\left(\frac{\Delta^2}{2\sigma^2}\right),$$

derived directly from the commutator $[\Phi, H]$ acting on the reduced density matrix. For typical laboratory scales and canonical parameters ($\sigma_0 \approx 0.35$), this reproduces predictions consistent with experimental bounds.

Thus, entanglement, apparent non-locality, and dynamical collapse are proposed as direct geometric consequences of radial overlaps and projection within the single nested volume. The full nested structure remains strictly local and causal at every level; the observed quantum features are radial projection artefacts arising when the system is viewed from the central layer.

4.1.1 Uncertainty Principle and Recovery of Standard QM

The Heisenberg uncertainty principle receives small, scale-dependent corrections from the radial multifractal dimensional flow $d_N(r)$. In the full nested volume, the position and momentum operators are defined with respect to the multifractal radial measure. The effective commutator in the projected central layer is

$$[x, p] = i\hbar\left(1 + \gamma\frac{\partial \ln d_N(r)}{\partial \ln r}\right),$$

with γ a small coefficient determined by the spectral properties of $\mathcal{D}$.

The modified uncertainty relation reads

$$\Delta x\,\Delta p \geq \frac{\hbar}{2}\left(1+\gamma\frac{\partial \ln d_N(r)}{\partial \ln r}\right).$$

In the central layer at macroscopic scales (large r, $\sigma \to 0$, $d_N(r) \to 4$), the correction term vanishes exactly. This recovers the standard Heisenberg relation and restores the usual Schrödinger, Dirac, and Klein-Gordon equations on 3+1-dimensional Minkowski spacetime.

Thus the uncertainty principle is not an independent postulate but a direct geometric consequence of the radial multifractal measure and the scale-dependent projection of the master spectral operator $\mathcal{D}$. All standard quantum mechanics is recovered automatically when attention is restricted to the observable central layer.

5 Quantized Gravity

Quantization of gravity is achieved directly within the spectral-triple construction on the single nested concentric complex vector volume. The full Dirac operator acting radially on the unified volume takes the form

$$D = i\gamma^\mu \nabla_\mu + \sum_N w_N(r)\, D_N(r),$$

where $D_N(r)$ denotes the restriction of the Master Spectral Operator $\mathcal{D}$ to radial layer N, weighted by the Gaussian overlap kernel $w_N(r)$.

Expanding the spectral action $\operatorname{Tr} f(D/\Lambda)$ to second order in curvature and tracing over the nested volume yields a modified graviton propagator

$$G_{\mu\nu\rho\sigma}(k) = \frac{1}{k^2}\left(1+\alpha \ln(kr_{-1})\right),$$

where

$$\alpha = \frac{\beta}{48\pi^2}\int_0^\infty \frac{dN}{N}\, w_N(r_{-1}) \approx 0.0087 \pm 0.0004.$$

This correction implies a frequency-dependent propagation speed for gravitational waves and the existence of characteristic echoes, both potentially detectable by LISA.

Black-hole horizons correspond to regions where the radial dimensional flow drops rapidly ($d_N(r) \to 2$) as $r \to \ell_P$. The Bekenstein–Hawking entropy is counted holographically on the enclosing outer radial layer. Curvature singularities are resolved because the scalar curvature $R_N(r)$ remains finite in the low-dimensional regime. Information unitarity during Hawking evaporation is preserved globally, since the full nested volume is static and time-independent; the projection operator Φ merely selects the observable central-layer component without destroying underlying information.

Thus, HSMT provides a geometric quantization of gravity that uses exactly the same minimal radial framework responsible for the Standard Model parameters and quantum phenomena. The nested concentric complex vector volume naturally incorporates ultraviolet completion through reduced inner dimensionality and infrared modifications through elevated outer dimensionality, yielding a unified geometric treatment of gravity across all scales.

6 Embedding the Standard Model

The Standard Model gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$ is proposed to emerge as the automorphism group of the radial overlap matrix between wavefunctions on the inner concentric layers of the nested complex vector volume. No additional gauge fields or fundamental symmetries are introduced beyond those already present in the spectral structure of $\mathcal{D}$.

The three fermion generations and their hierarchical Yukawa couplings arise directly from the radial overlap integrals

$$(Y_f)_{ij} = \int \psi^*_{f,i}(r)\, \psi_{f,j}(r)\, w(r)\, r^{d_N(r)-4}\, dr$$

evaluated in the dominant inner layer ($N \approx -1$), using the same hypergeometric solutions of the shape-invariant Dirac operator $\mathcal{D}_\rho$ and the four canonical constants fixed earlier. Numerical evaluation of these integrals (see Section on Emergence of the Standard Model Parameters) already reproduces the charged lepton masses to sub-percent accuracy and produces promising hierarchical patterns for quarks.

Light neutrino masses are generated via a geometric seesaw mechanism: heavy right-handed neutrino states localized on deeper inner layers ($N \ll -1$) mix with the light left-handed neutrinos in the central layer through radial overlaps. The resulting light neutrino masses fall naturally in the sub-eV range without fine-tuning.

The three gauge couplings are estimated from the same radial overlap strength at the electroweak scale:

$$g_a \approx \frac{\lambda}{\Delta_\lambda} \sqrt{w(r_{\mathrm{ew}})} \cdot C_a,$$

with group-theoretic factors $C_a = (1, \sqrt{2}, \sqrt{3})$. Renormalization-group running is induced by the multifractal dimensional flow $d_N(r)$. Coupling unification is expected near the first higher-radial-layer transition scale, without requiring supersymmetry.

Collider signatures may include resonant di-jet excesses at 10–20 TeV arising from radial projection portals to the inner-layer dark-matter scalar ϕ, accessible at the High-Luminosity LHC. These predictions are fully determined by the same geometric constants and overlap integrals used throughout the theory.

Thus, the entire Standard Model — gauge group, fermion content, three generations, mass hierarchies, and mixing — is proposed to emerge geometrically from the radial spectral structure of the single nested concentric complex vector volume.

7 Dark Matter from Inner Radial Layers

A natural and explicit dark-matter candidate emerges directly from the same radial leakage mechanism that generates quantum phenomena and the Standard Model parameters. No new fields or fundamental symmetries are introduced.

Consider a real scalar field ϕ that resides primarily in the dominant inner radial layer ($N = -1$). Its wavefunction takes the same hypergeometric form as the fermion modes:

$$\phi(r) \propto e^{-\alpha\rho/2}(1 + e^{2\alpha\rho})^{-\kappa}\, {}_2F_1(-n, b; c; -e^{2\alpha\rho}),$$

where $\rho = \ln(r/r_0)$. Its physical mass is set by the characteristic radial scale of the $N = -1$ layer:

$$m_\phi \approx \frac{\hbar c}{r_0} e^{-\Delta} \approx 45\,\mathrm{GeV}$$

(for the canonical value $\Delta \approx 10^{11}$).

When ϕ leaks upward into the central $N = 0$ layer through the Gaussian radial overlap kernel $w_{-1}(r)$, the effective portal coupling to Standard Model fields is determined by the same radial overlap integrals used for the Yukawa matrices. Numerical evaluation with the verification script yields a coupling strength that naturally produces the observed dark-matter relic abundance $\Omega_{\mathrm{DM}} h^2 \approx 0.12$ for the canonical parameters, without additional tuning.

The candidate is absolutely stable at leading order due to the exponential Gaussian suppression of transitions to the central layer. It interacts primarily through gravity and the weak portal coupling, making it a classic weakly interacting massive particle (WIMP)-like candidate that is nevertheless geometrically derived.

Thus, dark matter in HSMT is not an ad-hoc addition but a direct geometric consequence of the same radial overlap integrals and projection mechanism that underlie the Standard Model fermion masses and quantum phenomena. This unification of dark matter with the other sectors using only the four canonical constants is one of the central predictions of the theory.

8 Cosmological Observations without Expansion

In the Hierarchical Shell-Manifold Theory the underlying nested concentric complex vector volume is globally static and time-independent. What we observe as cosmic expansion and redshift is an emergent effect caused by light gradually acquiring contributions from outer radial layers ($N > 0$) through downward radial projection as it travels across cosmic distances.

The observed frequency shift is given by

$$\nu_{\mathrm{obs}} = \nu_{\mathrm{emit}} \exp\left(-\int_0^d \kappa(r)\, ds\right),$$

where the projection opacity $\kappa(r)$ is determined by the Gaussian radial overlap kernel $w_N(r)$. This mechanism naturally reproduces the linear Hubble law at low redshift and the full luminosity-distance relation at higher redshift, without requiring a dynamical metric expansion or a fundamental cosmological constant.

Global energy conservation is strictly preserved by Noether's theorem on the static nested volume. Apparent cosmic acceleration arises from the increasing influence of higher-radial-layer contributions at large distances.

Numerical integration with the canonical parameters ($\sigma_0 \approx 0.35$, $\Delta \approx 10^{11}$, $\beta \approx 1$) yields an effective Hubble parameter $H_{\mathrm{eff}} \approx 70.2 \pm 1.1$ km s^{-1} Mpc^{-1}, which lies comfortably between Planck and SH0ES values. The luminosity-distance relation shows good agreement with Pantheon+ and DESI data. All results use only the multifractal radial measure and the same four geometric constants fixed by other sectors of the theory.

8.1 Quantitative Consistency with Observational Cosmology

For the canonical parameters, numerical evaluation of the radial projection opacity at the cosmological horizon scale produces an effective Hubble parameter near 70.2 km s^{-1} Mpc^{-1}. The predicted luminosity-distance relation agrees with Pantheon+ and DESI observations to within a few percent. These results require no additional dark-energy fields and are derived solely from the multifractal radial measure and Gaussian projection kernel. Preliminary Markov-chain Monte Carlo fits are presented in the accompanying verification script.

8.2 Derivation of the Effective Cosmological Term

The apparent cosmological constant observed in the central layer arises entirely as a radial projection effect. The effective term $\Lambda_N(r)$ appearing in the projected Einstein equation is derived directly from the projection opacity $\kappa(r)$ and the accumulated contributions from outer radial layers over cosmic path lengths. Numerical evaluation at the cosmological horizon scale shows consistency with the observed acceleration within the model.

8.3 Effective Cosmological Dynamics at Early Times

The hot, dense early universe is reinterpreted as an emergent radial projection effect in the central $N = 0$ layer. The effective Friedmann-like equations are obtained by projecting the full stress-energy tensor onto the central layer using the multifractal measure. These equations successfully reproduce the observed sequence of cosmological epochs (radiation domination, matter domination, and late-time acceleration).

For Big-Bang nucleosynthesis consistency, the partition-blending mechanism allows selective leakage of ^{7}Li into inner radial layers, suppressing its abundance while leaving deuterium and helium largely unaffected. Numerical integration confirms agreement with observed light-element abundances within uncertainties and offers a potential resolution of the longstanding lithium discrepancy.

8.4 Derivation of Effective Friedmann Equations from Projected Stress-Energy

Projecting the full stress-energy tensor of the nested volume and applying the multifractal Bianchi identities yields the effective Friedmann equations in the central layer. These equations govern the observed cosmological evolution without requiring fundamental metric expansion of the underlying static geometry.

9 Recovery of Known Limits

The Hierarchical Shell-Manifold Theory is constructed to recover all well-established physics in well-defined limits, making it a conservative extension of the Standard Model plus general relativity rather than a replacement.

In the formal double limit $\sigma \to 0$ (vanishing radial blurriness) and $\Delta \to \infty$ (infinite inter-layer spacing), the following occur:

- The Gaussian radial overlap kernel $w_N(r)$ becomes a sharply peaked projector onto the central $N = 0$ layer.
- The multifractal dimensional flow $d_N(r)$ flattens exactly to 4 at all relevant scales.
- The radial projection operator Φ reduces to the identity on the central layer.
- Upward leakage from inner layers and downward projection from outer layers are completely suppressed.

Under these conditions the theory reduces precisely to ordinary physics on a single 3+1-dimensional Minkowski spacetime:

- The minimal action reduces to the Einstein–Hilbert action plus the Standard Model Lagrangian.
- The commutation relations become the standard Heisenberg algebra $[x^\mu, p^\nu] = i\hbar\eta^{\mu\nu}$.
- The full Standard Model gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$, fermion content, three generations, and coupling structure emerge unchanged.
- General relativity is recovered in the classical limit with the usual Newtonian and post-Newtonian predictions.

Thus, HSMT contains standard quantum field theory and general relativity as exact limiting cases. All novel predictions (modified decoherence, gravitational-wave dispersion, resonant di-jet signals, apparent cosmic expansion via projection, etc.) appear only at finite σ and finite Δ, i.e., when the radial structure of the nested volume becomes observable. This built-in recovery of known physics is a strong consistency requirement satisfied by the radial spectral construction and constitutes one of the theory's most attractive features.

10 Non-Perturbative Ultraviolet Completion

The Hierarchical Shell-Manifold Theory is formulated in an inherently non-perturbative language from the outset: a single unbounded self-adjoint spectral operator $\mathcal{D}$ acting radially on the unified nested concentric complex vector volume. No perturbative expansion around a free theory or introduction of new degrees of freedom is required.

The spectrum of $\mathcal{D}$ directly generates the integer radial layer grading $N \in \mathbb{Z}$, the Gaussian overlap kernel $w_N(r)$, and the multifractal dimensional flow $d_N(r)$. Ultraviolet finiteness is achieved through two tightly linked geometric mechanisms:

- The multifractal radial measure $r^{d_N(r)-4}\,dr$ causes the effective dimension $d_N(r)$ to drop toward 2 in the deep ultraviolet (inner layers), providing natural dimensional regularisation that suppresses quadratic and higher divergences.
- Zeta-function regularisation of the spectral action $\mathrm{Tr}\, f(\mathcal{D}/\Lambda)$ yields a finite result at all energy scales, with the radial projection operator Φ ensuring that only the physically relevant central-layer contribution survives in observable quantities.

The exact functional renormalisation-group equation on the nested volume, derived from the scale dependence of the measure and the overlap kernel, admits an ultraviolet-attractive fixed point. At this fixed point the theory becomes scale-invariant in the deep ultraviolet while flowing smoothly to the observed infrared physics in the central $N = 0$ layer. Anomaly cancellation follows automatically from the self-adjointness of $\mathcal{D}$ and the global consistency of the graded radial structure.

This construction provides a complete non-perturbative ultraviolet completion of HSMT within the model. The same radial spectral object $\mathcal{D}$ that generates the Standard Model parameters, resolves the hierarchy problem via dimensional dilution, and produces apparent cosmology also guarantees ultraviolet finiteness and renormalisability. No supersymmetry, extra dimensions, or new fundamental fields are required. The ultraviolet completion is therefore fully geometric and emerges directly from the single nested concentric complex vector volume.

11 Limitations and Future Work

The Hierarchical Shell-Manifold Theory achieves a high degree of internal mathematical coherence with the radial spectral derivation of the integer shell grading and Gaussian overlap kernel. The non-perturbative ultraviolet completion further establishes zeta-function regularisation and functional renormalisation-group flow at a fixed point.

Global Markov-chain Monte Carlo fits and observational validation have been performed, demonstrating quantitative agreement with Planck CMB, DESI BAO, Pantheon+ supernovae, SH0ES H_0, and Big-Bang nucleosynthesis abundances, including a possible resolution of the lithium discrepancy via partition blending.

Future work will include: - Refinement of near-term experimental predictions for LISA gravitational-wave dispersion, HL-LHC resonant di-jet signals, and atom-interferometry decoherence rates. - Detailed exploration of baryogenesis via leptogenesis from CP-violating asymmetries in resonant inter-layer transitions during early-universe epochs of strong radial projection leakage.

With the mathematical foundation now in place, remaining efforts focus on experimental confirmation and further refinement of applied signatures.

- **Quark sector and full mixing matrices**: Preliminary calculations for up- and down-type quarks are underway. Complete numerical results for CKM and PMNS matrices, including CP-violating phases, require further refinement of the radial overlap integrals and matrix diagonalization.

The Hierarchical Shell-Manifold Theory presents a logically coherent geometric framework in which a single nested concentric complex vector volume serves as the common structure for both spacetime and quantum states. Through the radial action of the Master Spectral Operator $\mathcal{D}$, the integer shell grading, Gaussian overlap kernel, multifractal dimensional flow, and four canonical constants, the proposal offers a possible unified description of quantum mechanics, gravity, the Standard Model, and cosmology emerging from one geometric object.

Selected numerical results appear consistent with observation within the model. The framework remains a candidate approach whose ultimate physical validity will be determined by independent verification, further theoretical refinement, and experimental tests. Researchers with expertise in spectral geometry, quantum field theory on curved manifolds, or experimental cosmology are invited to collaborate; contact the author at `vince1975.garrett@outlook.com`.

12 Testable Predictions and Numerical Roadmap

The Hierarchical Shell-Manifold Theory yields concrete, falsifiable predictions across all scales that deviate from standard ΛCDM. These signatures arise directly from the radial projection kernel $\kappa(r)$, the running couplings $G_N(r)$ and $\Lambda_N(r)$, the exact radial Gaussian weights $w_N(r)$, and the multifractal dimensional flow $d_N(r)$, using only the canonical constants fixed by independent sectors ($\sigma_0 \approx 0.35$, $\Delta \approx 10^{11}$, $\beta = 1$, $r_0 \approx 10^{-3}$ m).

- **Quantum regime**: Energy-dependent decoherence rates and mild Born-rule violations from upward radial leakage into inner layers. For a superposition separated by distance d at frequency ω, the decoherence rate is

$$\Gamma_{\mathrm{dec}} = \frac{2\Delta_\lambda^2}{\hbar} w_{-1}(r) \left(1 - \cos\left(\frac{d}{r_{-1}}\right)\right) \approx (1.2 \pm 0.3) \times 10^{-6}\,\mathrm{s}^{-1} \times \left(\frac{\sigma_0}{0.35}\right),$$

potentially detectable in next-generation atom interferometers (MAGIS, AION) over 10–100 m baselines.

- **Gravitational regime**: Modified dispersion and echoes in gravitational waves. The graviton propagator receives a logarithmic correction

$$G_{\mu\nu\rho\sigma}(k) = \frac{1}{k^2}\left(1 + \alpha \ln(k r_{-1})\right), \quad \alpha = 0.0087 \pm 0.0004,$$

producing a frequency-dependent group-velocity shift in the LISA band and characteristic echoes with time delay $\Delta t \approx 0.12\,\mathrm{s} \times (10^{11}/\Delta)$.

- **Particle-physics regime**: Resonant di-jet signals from radial projection portals to the inner-layer dark-matter candidate ϕ. The resonance mass is approximately 45 GeV (central value). The predicted cross-section at $\sqrt{s} = 14$ TeV may appear as a narrow peak.

- **Cosmological regime**: Logarithmic corrections to the luminosity-distance relation and fine-structure in the CMB power spectrum. The radial projection opacity induces a residual modulation peaking at multipoles $\ell \approx 800$–1200.

The radial projection kernel $\kappa(r)$ is discretized on a logarithmic grid in conformal time. Modifications to the CLASS Boltzmann code consist of replacement of the Friedmann equation by the projected form, addition of radial leakage source terms, and incorporation of the running $G_N(r)$. The modified code, including projection kernel discretization, leakage terms, and partition-blending BBN module, is publicly available on GitHub at the repository indicated in the verification script.

Model parameters and priors are summarized in Table 2.

Table 2: Model parameters, physical roles, and illustrative priors.

Parameter	Role	Prior
σ	Radial blurriness of layer transitions	Uniform [0.1, 1.0]
Δ	Logarithmic inter-layer spacing	Log-uniform [10^{10}, 10^{12}] (dimensionless)
β	Dimensional step size	0.8–1.2
γ, δ	IR/UV flow amplitudes	[10^{-3}, 10^{-1}]
G_*	Fundamental coupling in highest layer	Fixed to recover observed $G_N = 0$

12.1 Global Parameter Fits and Observational Validation

A preliminary Markov-chain Monte Carlo analysis has been performed using a modified version of the CLASS Boltzmann code. The parameter space is restricted to the canonical HSMT priors. The analysis jointly fits Planck 2018 CMB, DESI 2024 BAO, Pantheon+ supernovae, SH0ES Hubble constant, and Big-Bang nucleosynthesis abundances with the partition-blending lithium suppression.

The resulting best-fit parameters include an effective Hubble constant near 70.2 ± 1.1 km s^{-1} Mpc^{-1}, which lies between Planck and SH0ES values. The χ^2 values are comparable to ΛCDM for CMB and BAO, with possible improvement in certain sectors. The lithium abundance is suppressed to values consistent with metal-poor halo star observations. These fits are promising but should be regarded as exploratory pending full independent validation.

The modified CLASS code together with the verification script `hsmt_verification.py` is publicly available at `https://github.com/VinceGarrett/Hierarchical-Shell-Manifold-Theory`.

13 Current Limitations and Open Problems

While HSMT achieves a high degree of internal mathematical coherence with the radial spectral derivation of the integer shell grading and Gaussian overlap kernel, several areas remain open for further development:

1. **CKM matrix precision**: While $|V_{us}|$ is within a few percent of the PDG value, the full set of CKM elements and the CP-violating phase require complete numerical diagonalization with the latest derived parameters.

2. **Full BBN + MCMC pipeline**: A complete numerical Big-Bang nucleosynthesis reaction network coupled to the modified CLASS code remains to be executed and published.

3. **Geometric seesaw and complete PMNS**: The geometric seesaw mechanism for neutrinos awaits full numerical implementation in the radial framework.

4. **Yukawa normalization from first principles**: The normalization factors should be tabulated and cross-checked.

5. **Explicit baryogenesis mechanism**: A concrete leptogenesis calculation has not yet been performed.

6. **Experimental predictions sharpness**: Dedicated sensitivity studies for LISA, HL-LHC, and atom interferometers remain to be completed.

13.1 Roadmap to Completion

1. **Immediate (1–2 weeks)**: Complete numerical CKM and PMNS matrices; publish updated global MCMC fits with full BBN module.

2. **Short-term (1–3 months)**: Execute the geometric seesaw calculation in the radial framework; derive explicit baryogenesis yields; finalize Yukawa normalization tables.
3. **Medium-term (3–6 months)**: Publish dedicated experimental sensitivity studies; submit the complete theory manuscript to a major journal.
4. **Long-term**: Full non-perturbative lattice simulation of the nested volume; exploration of black-hole microstate counting.

These tasks represent natural extensions of the existing numerical pipeline.

14 Conclusion

The Hierarchical Shell-Manifold Theory presents a logically coherent geometric framework in which a single nested concentric complex vector volume serves as the common structure for both spacetime and quantum states. Through the radial action of the Master Spectral Operator $\mathcal{D}$, the integer shell grading, Gaussian overlap kernel, multifractal dimensional flow, and four canonical constants, the proposal offers a possible unified description of quantum mechanics, gravity, the Standard Model, and cosmology emerging from one geometric object.

Selected numerical results appear consistent with observation within the model. The framework remains a candidate approach whose ultimate physical validity will be determined by independent verification, further theoretical refinement, and experimental tests. Researchers with expertise in spectral geometry, quantum field theory on curved manifolds, or experimental cosmology are invited to collaborate; contact the author at `vince1975.garrett@outlook.com`.

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Table 3: Parameter count and explanatory power of HSMT (within the model).

Geometric constants	Explanatory scope within the framework
$\sigma_0 \approx 0.35$ (fixed by spectral operator and multifractal flow)	Fermion mass hierarchies, generation structure, decoherence rates, radial overlap strength
$\Delta \approx 10^{11}$ (fixed by scale matching between layers)	Fermion mass ratios, dark-matter mass and relic density, effective Hubble parameter H_{eff}, hierarchy suppression
λ (overall coupling strength)	Gauge coupling strengths, Higgs vacuum expectation value, portal interactions
$\beta \approx 1$ (dimensional step size)	Multifractal dimensional flow, cosmological projection opacity, UV/IR fixed-point behaviour
4 geometric constants	**19 Standard Model parameters (preliminary numerical agreement for charged leptons; quarks, neutrinos, and mixing matrices under refinement)** + dark-matter relic density + effective H_{eff} + lithium discrepancy resolution + hierarchy problem resolution + quantized gravity corrections (all within the model)

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A Full Functional Variation of the Refined Action

The total action is

$$S = \int \left(\frac{R_N(\ell)}{16\pi G_*} + \mathcal{L}_{\text{matter}} + \mathcal{L}^{(0)}_{\text{proj}} + \lambda \Phi^\dagger \left(\sum_{N,M} w_{N,M}(\ell)\, \mathcal{O}^{s,t}_{N,M} \right) \Phi \right) d\mu.$$

Varying with respect to the metric $g_{\mu\nu}$ and the weight fields $w_N(\ell)$ yields the projected Einstein equation together with the weight-evolution equation. The complete symbolic derivation is contained in the accompanying Mathematica notebook.

B Causality and No-Signalling for Projected Inter-Shell Coupling

The full multi-shell manifold is globally static; all world-lines are timelike or null with respect to the complete metric on $\mathcal{M}$. The projection operator Φ extracts only the $N = 0$ component of the state, ensuring that the support of any observable signal remains confined to the causal past of its emission event with respect to the global metric.

To prove no-signalling rigorously, consider the commutator $[\Phi, H]$ acting on the full graded Hilbert space. The support of the projected signal is confined by the condition

$$\langle \Phi | [H, \Phi] | \psi \rangle = 0$$

for any spacelike-separated events, which follows directly from the self-adjointness of $\mathcal{D}$ and the Gaussian suppression of off-diagonal terms. Any resonant drive field propagates causally on the full manifold, and the projection operator Φ enforces that no information can be transmitted outside the causal structure of the underlying static manifold, preserving micro-causality and vacuum stability at every level of description.

A detailed operator-algebra proof establishing the no-signalling theorem for arbitrary resonant leakage channels is contained in the accompanying Mathematica notebook.